

# Motion Timeline Editor

Keyframe editor with editable bezier easing curves, onion-skinning, and export to Lottie JSON or raw CSS keyframe animations.

|       |        |        |         |
|-------|--------|--------|---------|
| React | Canvas | Lottie | Zustand |
|-------|--------|--------|---------|

| VERSION | STATUS      | STAGE          | EXPORTS     | OWNER        | UPDATED  |
|---------|-------------|----------------|-------------|--------------|----------|
| 0.4.2   | In Progress | Internal alpha | Lottie, CSS | Studio Tools | Sep 2026 |

## 1. Overview

The Motion Timeline Editor is a keyframe animation tool for interface motion. Properties are animated on a timeline, easing between keyframes is edited as a cubic bezier curve directly on the canvas, and onion-skinning shows neighbouring frames as ghosts so spacing can be judged by eye rather than by scrubbing back and forth.

Output targets the two formats motion actually ships in: Lottie JSON for animations rendered by a player, and raw CSS keyframes for effects that belong in the stylesheet. The editor is explicit about which features survive each export, because a curve that looks right in the tool and degrades silently in CSS is worse than one the tool refuses.

## 2. Editing model

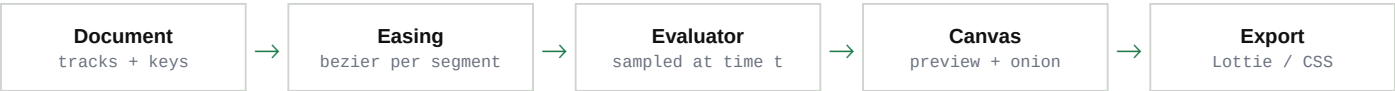


Figure 1 — One evaluator serves preview and export, so the exported animation is sampled from the same function the editor draws.

The document is a plain serialisable object held in a Zustand store, and every edit is an operation applied to it. That gives undo and redo for free, makes the file format identical to the in-memory model, and allows a change to be replayed against a document for regression testing of the exporters.

## 3. Components

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| Document model | Layers, tracks, keyframes, and markers in one immutable tree with an operation log. |
| Timeline UI    | Track rows, snapping, multi-select, and drag with modifier-based constraints.       |
| Easing editor  | Draggable cubic bezier handles per segment, with presets and numeric entry.         |
| Evaluator      | Deterministic property sampling at arbitrary time, shared by preview and exporters. |
| Onion skin     | Renders N frames before and after at reduced opacity, with a configurable step.     |
| Exporters      | Lottie JSON writer and CSS keyframe emitter, each with a capability check.          |

## 4. Document format

```
{
  "version": "0.4",
  "duration": 1800,           // ms
  "fps": 60,
  "layers": [{
    "id": "card", "type": "transform",
    "tracks": {
      "y": [
        { "t": 0,   "v": 24, "out": [0.33, 0.0] },
        { "t": 420, "v": 0,  "in": [0.18, 1.0] } // cubic bezier handles
      ],
      "opacity": [
        { "t": 0,   "v": 0 },
        { "t": 240, "v": 1, "in": [0.4, 1.0] }
      ]
    }
  }],
  "markers": [{ "t": 420, "label": "settle" }]
}
```

Listing 1 — Keyframes carry their own outgoing and incoming handles, so easing is per segment.

## 5. Export fidelity

| FEATURE                  | LOTTIE | CSS KEYFRAMES                                     |
|--------------------------|--------|---------------------------------------------------|
| Per-segment cubic bezier | Exact  | Exact, one animation per property.                |
| Transform and opacity    | Exact  | Exact.                                            |
| Colour interpolation     | Exact  | Exact in sRGB; wide-gamut approximated.           |
| Masks and path morphs    | Exact  | Unsupported; export blocked with the layer named. |

## 6. Current status

- **Working.** Timeline editing, bezier easing, onion-skinning, undo and redo, and Lottie export for transform, opacity, and colour tracks. In daily internal use for interface motion specs.
- **In progress.** CSS exporter handles single-property animations correctly but does not yet compose multi-property timelines into a single rule set; path morph support is unimplemented.
- **Not started.** Collaborative editing, asset import beyond SVG, and a plugin surface for custom property types.

## 7. Known limitations

- Documents above roughly 400 keyframes show interaction lag during drag, traced to full-tree re-render on each pointer move rather than to the evaluator.
- Onion-skin rendering is CPU-bound and drops to a 2-frame window automatically above 1440p.
- The Lottie writer targets a subset of the specification; files are validated against a player on export, and an unsupported construct fails loudly rather than producing a silent no-op.

## 8. Remaining work

| MILESTONE  | SCOPE                                                                   | TARGET  |
|------------|-------------------------------------------------------------------------|---------|
| Alpha exit | Multi-property CSS export, drag performance at 1,000 keyframes.         | Q4 2026 |
| Beta       | Path morphing, SVG asset import, shareable read-only preview links.     | Q1 2027 |
| 1.0        | Stable document format, plugin surface, and exporter conformance suite. | Q2 2027 |